

Clinical Scenario — Female, Age 38 · Opioid ED Cohort (25–44 band) · Density Bin: *high* (14 claims in 30-day window)

Active prescriptions: Oxycodone HCl 30 mg/day (90-day supply) · Gabapentin 1,800 mg/day · Alprazolam 1 mg TID

Diagnoses: M54.5 Low back pain · G89.29 Other chronic pain · F41.1 Anxiety disorder · Z79.891 Long-term opioid use

Flags: No physical therapy referral in 12-month lookback · 3 opioid refills in 90 days · Opioid + benzodiazepine co-prescription (FDA black-box warning 2016)

(1) OBSERVE

Tab 1–2 · Risk Assessment

Risk score: **0.87**
Risk band: **HIGH**

Model agreement:
XGBoost: 0.85
CatBoost: 0.89
Ensemble: 0.87

Density-bin routing:
high stratum
(14 claims)

Holdout AUROC: 0.979
Holdout PR-AUC: 0.935

Action trigger:
Score > 0.75 threshold
→ Orient workflow
auto-launched

(2) ORIENT

Tab 4 · Causal & Trajectories

SHAP top features:

- gabapentin_count
(mean |SHAP| = 0.34)
- chronic_pain_dx
(0.28)
- pgx_num_cplic_drugs
(0.22, CYP2D6 signal)

DTW archetype:
Chronic-Escalation
Month 9 of 22.1 avg.
3 windows remain

FP-Growth:
Oxy+Gaba+Alp triple
present in 67% of
high-risk cases (lift 4.1)

(3) DECIDE

Tab 3 · What-If & CPIC

Counterfactual $\Delta\hat{p}$:
Add PT referral:
 $\Delta\hat{p} = -0.09 \rightarrow 0.78$

Taper gabapentin:
 $\Delta\hat{p} = -0.12 \rightarrow 0.75$

Both combined:
 $\Delta\hat{p} = -0.19 \rightarrow 0.68$

CPIC genotype lookup:
CYP2D6 proxy score
→ Poor Metabolizer
phenotype inferred

Oxycodone NOT recommended
(CPIC Level A)
Ultra-slow conversion
→ morphine accumulation

(4) ACT

Tab 5 · PGx Patient Card

CPIC recommendation:
CYP2D6 PM phenotype
Avoid oxycodone
Prefer hydromorphone
(CYP3A4 substrate;
no CYP2D6 reliance)

SNP pathway detail:
CYP2D6 *4/*4
(rs3892097, 1846G>A)
splice-site → zero enzyme
Oxy→oxymorphone:
analgesia fails +
parent Oxy accumulates
OPRM1 A118G
(rs1799971)
↓ μ -receptor affinity
compounds PM failure
CYP3A4 *1/*1
normal → std.
hydromorphone dose

30-day reassessment:
Expected score: **0.68**

PGx Intervention Outcome (30-day reassessment target)

- (1) **Drug switch:** Oxycodone (CYP2D6-metabolized, CPIC Level A caution for Poor Metabolizers) → Hydromorphone (CYP3A4 substrate; no CYP2D6 metabolism step; equianalgesic at reduced starting dose per CPIC guidance)
- (2) **Care pathway addition:** Physical therapy referral at Chronic-Escalation intervention window 2 of 3 (month 9; reduces opioid-refill transition probability by 34% per BupaR process map)
- (3) **Adjuvant taper:** Gabapentin 10%/week taper initiated (top Consensus-Causal feature, mean |SHAP| = 0.34; opioid+gabapentin co-prescription FDA black-box warning; IR rank #1 for 25–44 band)

Predicted risk after intervention: $\hat{p} = 0.68$ (MODERATE band) — **22-point reduction** from baseline $\hat{p} = 0.87$. Next PDMP review at 30 days; PGx Card downloadable for dispensing pharmacist.